

STATISTICS 652

LECTURE 1 - 26

o Discuss: Policy/Syllabus/Website

MOTIVATING EXAMPLES

1) EXPONENTIAL DENSITY: $f(x; \lambda) = \frac{1}{\lambda} e^{-x/\lambda}$ $\lambda > 0, x \geq 0$

RANDOM SAMPLE: $\underline{x} = (x_1, \dots, x_n)$

LIKELIHOOD: $L(\lambda; \underline{x}) = f(\underline{x}; \lambda) = \lambda^{-n} e^{-\bar{x}/\lambda}$ $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

LOG-LIKELIHOOD: $\ell(\lambda; \underline{x}) = -n \log \lambda - \frac{n}{\lambda} \bar{x}$

FIND MAXIMUM LIKELIHOOD ESTIMATE FOR λ

$\ell'(\lambda) = -\frac{n}{\lambda} + \frac{n}{\lambda^2} \bar{x} = 0 \Leftrightarrow \lambda = \bar{x}$ ONLY CRITICAL POINT

$\ell''(\lambda) = \frac{n}{\lambda^2} - \frac{2n}{\lambda^3} \bar{x} \begin{cases} < 0 & \lambda < 2\bar{x} \\ > 0 & \lambda > 2\bar{x} \end{cases}$ CONCAVE
CONVEX

$\Rightarrow \hat{\lambda} = \bar{x}$ IS LOCAL MAXIMUM Q: GLOBAL MAXIMUM? WHY?

$\Rightarrow \hat{\lambda} = \bar{x}$ IS MLE

2) REPARAMETERIZE: $\theta = 1/\lambda$ i.e. 1-1 $\Rightarrow f(x; \theta) = \theta e^{-\theta x}$ $\theta > 0, x \geq 0$

$\ell(\theta; \underline{x}) = n \log \theta - n \theta \bar{x}$

$\ell'(\theta) = \frac{n}{\theta} - n \bar{x} \Rightarrow \theta = \frac{1}{\bar{x}}$ ONLY CRITICAL POINT

$\ell''(\theta) = -\frac{n}{\theta^2} < 0 \Rightarrow \ell(\theta)$ is STRICTLY CONCAVE function $\Rightarrow \hat{\theta} = \frac{1}{\bar{x}}$ MLE.

RECALL: INVARIANCE OF MLE UNDER 1-1 REPARAMETERIZATION:

$\Rightarrow \hat{\lambda} = 1/\hat{\theta} = \bar{x}$

$$3) \text{ NORMAL } (\mu, \sigma^2) = (\mu, \theta) \quad \mu \in \mathbb{R}, \theta = \sigma^2 > 0$$

$$\underline{X} = (X_1, \dots, X_n) \text{ i.i.d. } N(\mu, \sigma^2)$$

$$\begin{aligned} \ell(\mu, \theta; \underline{x}) &= -\frac{n}{2} \log \theta - \frac{1}{2\theta} \sum_{i=1}^n (x_i - \mu)^2 - \frac{n}{2} \log(\sqrt{2\pi}) \\ &= -\frac{n}{2} \log \theta - \frac{n}{2\theta} (S^2 + (n-1)\bar{x}^2) - \frac{n}{2} \log(\sqrt{2\pi}) ; S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \end{aligned}$$

$$\partial \ell(\mu, \theta) / \partial \mu = \frac{n}{\theta} (\bar{x} - \mu)$$

$$\partial \ell(\mu, \theta) / \partial \theta = -\frac{n}{2\theta} + \frac{n}{2\theta^2} (S^2 + (n-1)\bar{x}^2)$$

$$\nabla \ell = \begin{pmatrix} \partial \ell / \partial \mu \\ \partial \ell / \partial \theta \end{pmatrix} = \underline{0} \iff \mu = \bar{x}, \theta = S^2 \quad \text{ONLY CRITICAL POINT}$$

$$\frac{\partial^2 \ell}{(\partial \mu)^2} = -\frac{n}{\theta} \quad \frac{\partial^2 \ell}{(\partial \theta)^2} = \frac{n}{2\theta^2} - \frac{n}{\theta^3} (S^2 + (n-1)\bar{x}^2) \quad \frac{\partial^2 \ell}{\partial \mu \partial \theta} = -\frac{n(\bar{x} - \mu)}{\theta^2}$$

$$\text{HESSIAN: } H = \begin{bmatrix} \partial^2 \ell / \partial \mu^2 & \partial^2 \ell / \partial \mu \partial \theta \\ \partial^2 \ell / \partial \mu \partial \theta & \partial^2 \ell / \partial \theta^2 \end{bmatrix} \neq 0 \quad \text{IN GENERAL} \\ \text{i.e. NOT CONCAVE FUNCTION}$$

$$\text{NOTE: } H < 0 \text{ NEGATIVE DEFINITE} \quad H \leq 0 \text{ NEGATIVE SEMI-DEFINITE}$$

$$H > 0 \text{ POSITIVE DEFINITE} \quad H \geq 0 \text{ POSITIVE SEMI-DEFINITE}$$

$$\text{AT } (\mu, \theta) = (\bar{x}, S^2) \quad H = \begin{bmatrix} -n/\theta & 0 \\ 0 & -n/(2\theta^2) \end{bmatrix} < 0 \Rightarrow \text{LOCAL MAXIMUM}$$

GLOBAL MAXIMUM: FOR differentiable $g: \mathbb{R}^2 \rightarrow \mathbb{R}$, $g \geq 2$, it

is possible for g to have ONLY ONE CRITICAL POINT, WITH THAT UNIQUE CRITICAL POINT BEING A LOCAL MAX BUT NOT THE GLOBAL MAX. (i.e., GLOBAL MAX MAY NOT EXIST. NEED TO EXAMINE BOUNDARIES: $(\mu, \theta) \rightarrow$

$$\circ \text{ PROOF OF GLOBAL MAX: } \ell(\mu, \theta) \leq \ell(\bar{x}, \theta) \quad (\pm \infty, [0, \infty))$$

$$\leq \ell(\bar{x}, S^2) \quad \text{USING ONE DIMENSIONAL ARGUMENTS}$$

$\circ$ REPARAMETERIZATION? TO STRICTLY CONCAVE FUNCTION?

EXPONENTIAL FAMILIES

DEF 1.1: A ONE-PARAMETER EXPONENTIAL FAMILY $\mathcal{P} = \{P_\theta / \theta \in \Theta\}$, P_θ PROBABILITY MEASURE ON $\mathbb{R}^m$, $\Theta \subset \mathbb{R}^p$, is an identifiable family of distributions with density or mass function OF THE FORM

$$f(x; \theta) = \exp \{ c(\theta) T(x) + d(\theta) + S(x) \} I_A(x)$$

where $c: \Theta \rightarrow \mathbb{R}$, $d: \Theta \rightarrow \mathbb{R}$, $T: \mathbb{R}^m \rightarrow \mathbb{R}$, $S: \mathbb{R}^m \rightarrow \mathbb{R}$ and with the set $A \subset \mathbb{R}^m$ NOT DEPENDENT ON θ . (REGULAR) (* OR ONE-DIMENSIONAL)

BRIEFLY DISCUSS: DENSITY = RADON-NIKODYN DERIVATIVE

(SINGULAR DISTRIBUTION, e.g. on a sphere)

EXAMPLES:

(1) $X \sim \text{EXP}(\lambda)$, i.e. $f(x; \lambda) = \frac{1}{\lambda} e^{-x/\lambda}$

$c(\lambda) = -1/\lambda$, $T(x) = x$, $S(x) = 0$, $d(\lambda) = -\log \lambda$, $A = \{x \geq 0\}$

OR $c(\lambda) = -(\frac{x}{\lambda} - 1)$, $T(x) = \frac{x}{\lambda} - 1$, $S(x) = -(\frac{x}{\lambda} - 1)$, $d(\lambda) = -\log \lambda - \frac{2}{\lambda}$, $A = \{x \geq 0\}$

i.e. NOT A UNIQUE REPRESENTATION

(2) $X \sim \text{BINOMIAL}(N, \theta)$ N -KNOWN $0 < \theta < 1$

$f(x; \theta) = \binom{N}{x} \theta^x (1-\theta)^{N-x}$ $x = 0, 1, \dots, N$

$= \exp \left\{ \left(\log \frac{\theta}{1-\theta} \right) x + N \log(1-\theta) + \log \binom{N}{x} \right\} I_A(x)$ $A = \{0, 1, \dots, N\}$

$\Rightarrow$ EXPONENTIAL FAMILY

NOTE: $c(\theta) = \log \left(\frac{\theta}{1-\theta} \right) = \log \eta(\theta)$ i.e. log-odds

$$(3) X \sim N(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma^2 > 0$$

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2/\sigma^2} \quad x \in \mathbb{R}$$

$$= \exp \left\{ -\frac{1}{2\sigma^2} x^2 + \frac{\mu}{\sigma^2} x - \frac{\mu^2}{2\sigma^2} - \log(\sigma \sqrt{2\pi}) \right\} \quad x \in \mathbb{R}$$

a) Fixed σ^2 : EXPONENTIAL FAMILY IN $\mu \in \mathbb{R}$

$$c(\mu) = \frac{\mu}{\sigma^2}, T(x) = x, S(\mu) = -\frac{\mu^2}{2\sigma^2}, d(\mu) = -\frac{\mu^2}{2\sigma^2} - \log(\sigma \sqrt{2\pi})$$

b) Fixed μ : EXPONENTIAL FAMILY IN $\sigma^2 > 0$

$$c(\sigma^2) = -\frac{1}{2\sigma^2}, T(x) = (x-\mu)^2, d(\sigma^2) = -\log(\sigma \sqrt{2\pi}), S(\mu) = 0$$

DEFINITION 1.2: A k -dimensional exponential family $\mathcal{P} = \{P_\theta \mid \theta \in \Theta\}$, $\Theta \subset \mathbb{R}^k$ is an identifiable, regular family of distributions with density or mass function of the form

$$f(x; \theta) = \exp \left\{ \sum_{j=1}^k c_j(\theta) T_j(x) + d(\theta) + S(x) \right\} I_A(x)$$

and k is the smallest integer for which such a representation holds.

EXAMPLE: $X \sim \text{NORMAL}(\mu, \sigma^2)$ is a 2-dimensional exponential family

$$\begin{aligned} c_1(\theta) &= -\frac{1}{2\sigma^2} & c_2(\theta) &= \frac{\mu}{\sigma^2} & d(\theta) &= -\frac{\mu^2}{2\sigma^2} - \log(\sigma \sqrt{2\pi}) \\ T_1(x) &= x^2 & T_2(x) &= x & S(x) &= 0 & A &= \mathbb{R} \end{aligned}$$

REMARK 2: c_1, c_2

P_θ is an exponential family in $\theta = (\theta_1, \theta_2)$

$\Rightarrow$ FOR FIXED θ_2 , IT IS AN EXP. FAMILY IN θ_1 , AND

FOR FIXED θ_1 , IT IS AN EXP. FAMILY IN θ_2 . (HW)

$\Leftarrow$ OPTIONAL HW

EXAMPLE:

$$X \sim \text{MULTINOMIAL}_m(N, \theta) \quad \theta = (\theta_1, \dots, \theta_m) \quad \sum_{j=1}^m \theta_j = 1, \quad 0 < \theta_j < 1$$

$$f(x; \theta) = \binom{N}{x_1, \dots, x_m} \theta_1^{x_1} \cdots \theta_m^{x_m} I_A(x)$$

$$\text{where } A = \{x \mid x_j = 0, 1, \dots, N \text{ for } j=1, \dots, m \text{ and } \sum_{j=1}^m x_j = N\}$$

$$f(x; \theta) = \exp \left\{ \sum_{j=1}^m x_j \log \theta_j + \log \binom{N}{x_1, \dots, x_m} \right\} I_A(x)$$

$$= \exp \left\{ \sum_{j=1}^{m-1} x_j \log (\theta_j / \theta_m) + N \log \theta_m + \log \binom{N}{x_1, \dots, x_m} \right\} I_A(x)$$

$\Rightarrow k = (m-1)$ PARAMETER/DIMENSIONAL EXPONENTIAL FAMILY

LEMMA 1.3. A representation for an exponential family corresponds to k -dimensions (k is the minimal representation) IFF

$$i) \sum_{j=1}^k a_j C_j(\theta) = a_0 \quad \forall \theta \in \Theta \Leftrightarrow a_0 = a_1 = \dots = a_k = 0, \quad \text{AND}$$

$$ii) \sum_{j=1}^k b_j T_j(x) = b_0 \quad \forall x \in A \Leftrightarrow b_0 = b_1 = \dots = b_k = 0.$$

i.e. THE FUNCTIONS $\{1, C_j(\theta), j=1, \dots, k\}$ are linearly independent
and the FUNCTIONS $\{1, T_j(x), j=1, \dots, k\}$ are linearly independent.

PROOF: OPTIONAL HW